AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

Type 1: Research Project

ReXTrust: XAI-Driven Multi-Disease Diagnosis in Chest X-rays via Conditional Enhancement and Grounded Reporting

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Change in Project Direction

Previous Idea

- Cognitive Load Estimation using STEW Dataset (EEG signals)
- Two-stage pipeline:
  - Model 1: Predict cognitive load from EEG data
  - Model 2: Predict cognitive load from video features
- Goal: Predict cognitive load of visual data

Challenges Identified

- Lack of dataset with paired EEG + video data
- Difficulty in mapping video features to cognitive load
- Limited scalability due to EEG dependency

Problem Statement

Chest X-ray diagnosis using AI suffers from three unresolved challenges:

- **Degraded Image Quality**
  X-rays from low-resource settings suffer from noise, low contrast, and poor illumination, reducing diagnostic reliability

- **Lack of Per-Disease Spatial Verification**
  Existing explainability methods produce blended saliency maps, making it difficult to localize individual diseases

- **Ungrounded AI-Generated Reports**
  Vision-language models generate fluent clinical text without verification, leading to potential hallucinated diagnoses

Literature Review

S.No	Paper Title	Objective	Model and Accuracy	Research Gaps
1	A Diffusion Model for Universal Medical Image Enhancement (UniMIE) (2025)	To develop a training-free universal model that enhances medical images across different modalities without needing paired data	Uses a pre-trained DDPM from ImageNet + guided diffusion (conditional denoising using input image) + degradation model (f, M)	High inference time due to patch-based diffusion scaling with image resolution.
2	All-in-One Medical Image Restoration with Latent Diffusion-Enhanced Vector-Quantized Codebook Prior (2025)	To develop a single unified model that can restore multiple medical imaging tasks by handling different types of degradation using a codebook prior	DiffCode – a unified model combining a VQ codebook bank, latent diffusion model, and task-aware routing (TARM) for multi-task medical image restoration.	Struggles with feature distribution misalignment between degraded (LQ) and clean (HQ) representations.
3	Score-CAM: Score-Weighted Visual Explanations for Convolutional Neural Networks(2025)	Gradient-free visual explanation for CNNs using increase-in-confidence scoring per activation map to generate maps	VGG16 on ImageNet (ILSVRC2012); Average Drop 31.5%, Average Increase 30.6%, Energy-Based Pointing Game 63.7%	Evaluated on single-label datasets only; authors note segmentation label evaluation is more accurate than bounding box but left for future work

Literature Review

S.No	Paper Title	Objective	Model and Accuracy	Research Gaps
4	FM-G-CAM: A Holistic Approach for Explainable AI in Computer Vision (2025)	Provide holistic visual explanation of CNN predictions by considering multiple top predicted classes in a single fused saliency map	ResNet50V2 on MS-COCO (80K+ images); FM-G-CAM outperforms Grad-CAM on IC and DC metrics across all top-5 class ranks	Fused output discards per-class spatial information at each pixel — individual disease regions cannot be verified independently.
5	Overview of Class Activation Maps for Visualization Explainability	Comprehensive survey of CAM-based XAI methods — gradient-based and gradient-free — with evaluation metrics and future research directions	Comparative ADCC scores — Grad-CAM 76.4, Score-CAM 78.55	Explicitly identifies multi-label scenarios with overlapping regions as an open research direction; notes existing metrics can be gamed — proposes ADCC as robust alternative
6	Universal Report Generation (UniRG-CXR): Scaling with Multimodal Reinforcement Learning (March 2025	To align model training with real-world radiology practice by optimizing clinically grounded rewards instead of standard text.	Generative VLM optimized via Multimodal RL using RadCliQ and LLM-based clinical error.Achieved SOTA across report-level metrics and disease-level diagnostic accuracy.	Primarily focuses on "clinical factuality" through reward signals but does not provide visual "receipts" (heatmaps) to prove that the generated text is derived from specific image regions

Literature Review

S.No	Paper Title	Objective	Model and Accuracy	Research Gaps
7	Agentic Medical Knowledge Graphs (AMG-RAG): Enhancing Medical QA (November 2025)	To automate the construction of dynamic Medical Knowledge Graphs (MKG) using LLM agents to mitigate hallucinations in complex medical reasoning	Agentic workflow using specialized LLM agents for autonomous search and graph building.Achieved 74.1% F1 on MEDQA and 66.34% accuracy on MEDMCQA.	While effective for medical QA, reasoning is limited to text retrieval and lacks pixel-level grounding for report generation.
8	MediVLM: A Vision Language Model for Radiology Report Generation (November 2025)	To generate radiology reports using an unsupervised setting and compute a "Severity Score" to prioritize urgent clinical cases.	Faster R-CNN + ClinicalBERT (Text Encoder) + GPT-2/BioT5 (Decoder). Accuracy: Achieved SOTA results on IU X-Ray and CASIA-CXR.	Relies on a fixed object detector for anatomical patches rather than disease heatmaps. Lacks explicit spatial grounding at the sentence level to prevent hallucinations in complex multi-label cases.
9	Densely Connected Convolutional Networks	To propose a CNN architecture that improves feature reuse and gradient flow through dense connectivity.	DenseNet is used Accuracy: Achieves SOTA on CIFAR-10/100 and SVHN; on ImageNet, matches ResNet performance with fewer parameters.	Does not address multi-label disease classification. No spatial localization or explainability mechanism. Evaluated only on general vision benchmarks with no clinical applicability.

Literature Review

S.No	Paper Title	Objective	Model and Accuracy	Research Gaps
10	CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Learning	To develop a deep learning model for automated chest X-ray disease detection and achieve performance exceeding practicing radiologists.	Model: DenseNet-121 with transfer learning (pretrained on ImageNet) trained on NIH ChestX-ray14. Sigmoid activation for multi-label classification. Weighted BCE loss for pneumonia task; Unweighted BCE for 14-class extension. Accuracy: Exceeded average radiologist performance on pneumonia detection (F1: 0.435 vs radiologist avg 0.387, statistically significant). Achieved SOTA AUROC on all 14 pathologies in ChestX-ray14.	Does not address multi-label disease classification. No spatial localization or explainability mechanism. Evaluated only on general vision benchmarks with no clinical applicability..

Literature Survey Summary

Recent advancements in chest X-ray analysis focus on image enhancement and disease detection.

Image Enhancement:

- Diffusion-based models such as UniMIE enable universal medical image enhancement [1].
- Improves image quality by addressing noise and low contrast.

Disease Detection:

- DenseNet-121 is widely used for multi-label classification [2].
- Sigmoid activation outputs confidence scores for multiple diseases.

Literature Survey Summary (Continued)

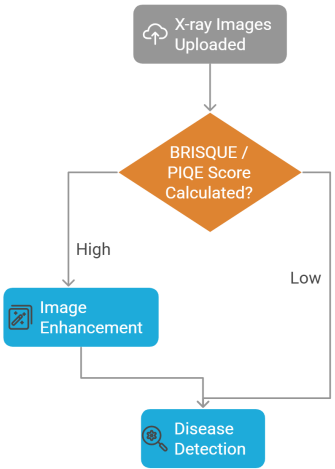
Explainable AI (XAI):

- Score-CAM generates class-specific saliency maps[3].
- Highlights spatial regions responsible for each disease prediction.

Vision-Language Models:

- MediVLM generates radiology reports from X-rays [4].
- AMG-RAG improves reliability using structured medical knowledge.

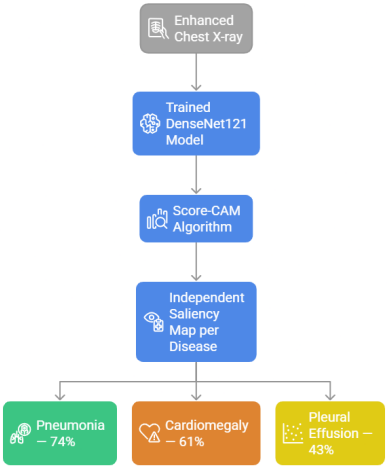
Novelty 1 – Conditional Image Enhancement



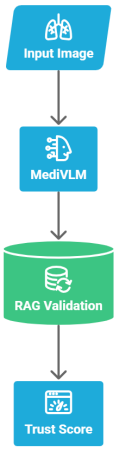
- Quality-aware preprocessing using BRISQUE / PIQE
- Decision-based pipeline: enhance only if quality is low
- UniMIE (diffusion model) for image enhancement[1]
- Training-free (no medical paired data required)
- Avoids unnecessary enhancement of good images
- Preserves clinical details while improving poor images

Novelty 2 – Multi-Disease Detection with Class-Specific Explainability

- Input: Preprocessed chest X-ray standardized for model inference
- DenseNet121 extracts deep spatial features from the image
- Final convolution layer produces activation maps capturing key patterns
- Score-CAM evaluates map importance using model confidence (no gradients required)
- Generates one saliency map per predicted disease (multi-label setting)
- Output: Disease probabilities along with visual explanations for interpretability



Novelty 3 – Trust-Aware Medical Report Generation using VLM + RAG

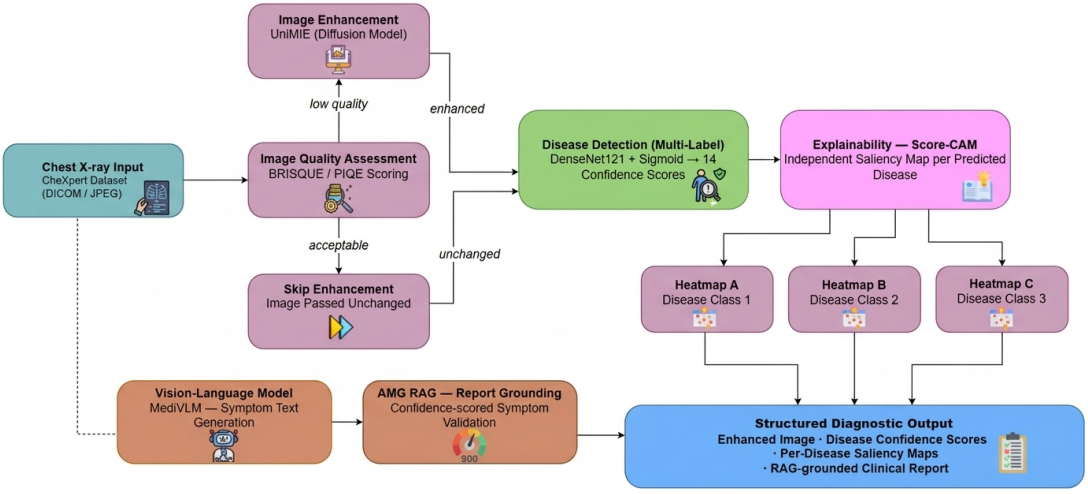


There is focal consolidation in the right lower lobe. Increased opacity is noted in the right lung base. The costophrenic angle is blunted on the right. Heart size is mildly enlarged. No pneumothorax is identified.

Finding in Text	MKG Validation	Confidence
Consolidation → RLL	Valid anatomical location	0.98
Opacity → lung base	Valid finding	0.96
Blunted CPA → right	Valid (effusion indicator)	0.97
Consolidation + Blunting → Pneumonia	Valid clinical relationship	0.94
No pneumothorax	Valid negative finding	0.99
Mild cardiomegaly	Valid separate finding	0.98

Overall Confidence = (0.98 + 0.96 + 0.97 + 0.94 + 0.99 + 0.98) ÷ 6 = 0.97

Architecture Diagram



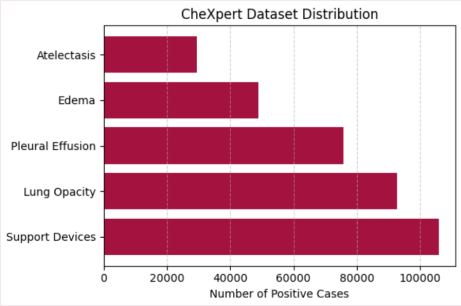
Dataset Description - CheXpert

224,316
X-ray Images

65,240
Patients

14
Observations

Class Distribution



Dataset Insights

- CheXpert [5] is a large-scale chest X-ray dataset from Stanford [2].
- Contains 224K images from 65K patients.
- Includes 14 disease labels with uncertainty annotations.
- Suitable for multi-label classification tasks.

References I

- B. Fei, Y. Li, W. Yang, H. Gao, J. Xu, L. Ma, Y. Yang, and P. Zhou, “A diffusion model for universal medical image enhancement,” *Communications Medicine*, vol. 5, p. 294, 2025.
- J. Irvin, P. Rajpurkar, M. Ko, Y. Yu, S. Ciurea-Ilcus, C. Chute, H. Marklund, B. Haghgoo, R. Ball, K. Shpanskaya *et al.*, “Chexpert: A large chest radiograph dataset with uncertainty labels and expert comparison,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2019, pp. 590–597.
- H. Wang, Z. Wang, M. Du, F. Yang, Z. Zhang, S. Ding, P. Mardziel, and X. Hu, “Score-cam: Score-weighted visual explanations for convolutional neural networks,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2020.
- D. Goswami, R. Subedi, and S. Chakraborty, “Medivlm: A vision language model for radiology report generation from medical images,” in *Findings of the Association for Computational Linguistics (EMNLP)*, 2025, pp. 10 287–10 304.

References II



Stanford AIMI, “Chexpert chest x-rays dataset,” https://aimi.stanford.edu/datasets/chexpert-chest-x-rays, 2019, accessed: 2026-03-30.

Thank You